

- * **Abnormal Colligative Properties** : When solute particle associated or dissociated in solvent.

$$i = \text{Vant Hoff factor} = \frac{\text{Actual moles of solute}}{\text{Moles of solute without dissociation or association}} = \frac{\text{Observed or experimental colligative properties}}{\text{Theoretical or calculated colligative properties}}$$

- (i) Relative lowering of vapour pressure : $\frac{P^0 - P}{P^0} = \frac{i \cdot n_{\text{solute}}}{i \cdot n_{\text{solute}} + n_{\text{solvent}}}$
 (ii) Elevation of boiling point : $\Delta T_b = i \cdot K_b \cdot m$
 (iii) Depression of freezing point : $\Delta T_f = i \cdot K_f \cdot m$
 (iv) Osmotic pressure : $\pi = i \cdot CRT$

For dissociation : $i = 1 + (n-1)\alpha$; For association : $i = 1 + \left(\frac{1}{n} - 1\right)\alpha$

Degree of dissociation $\alpha = \frac{i-1}{n-1}$; Degree of association $\alpha = \frac{1-i}{1-1/n}$

Where n = Total number of particles of solute after dissociation or association.

Henry's Law

This law deals with dissolution of gas in liquid i.e., mass of any gas dissolved in any solvent per unit volume is proportional to pressure of gas in equilibrium with liquid.

- (i) $m = k \times P$
 Where m = mass of gas dissolved per unit volume, k = proportionality constant, P = pressure of gas.
 (ii) $P = K_H \cdot X_{\text{gas}}$, Where P = Pressure of gas, K_H = Henry's constant, X_{gas} = Mole fraction of gas

LECTURE SHEET

EXERCISE-I

(Types of Solutions, Molarity, Normality, Molality, Mole fraction)

LEVEL-I (MAIN)

Straight Objective Type Questions

- If air is taken as a binary solution, the solvent is
 1) N_2 2) O_2 3) CO_2 4) H_2
- The physical change among the following is
 1) burning of coal 2) burning of sulphur
 3) dissolution of Glucose in water 4) burning of white phosphorus
- Homogeneous system among the following is
 1) milk 2) sand in water 3) urea in water 4) benzene in water
- The units of Molarity are
 1) gms. lit^{-1} 2) moles. lit^{-1} 3) equivalents. lit^{-1} 4) moles. kg^{-1}
- To half the molarity of a solution the following should be adopted
 1) weight of the solute to be doubled 2) weight of the solvent to be doubled
 3) volume of the solvent to be doubled 4) volume of the solution to be doubled

6. (A) : Molarity of a solution decreases with an increase of temperature.
(R) : As temperature increases volume of the solution increases.
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false
4) (A) is false but (R) is true
7. The units of Normality are
1) moles, lit^{-1} 2) moles, Kg^{-1} 3) equivalents, lit^{-1} 4) equivalents, Kg^{-1}
8. The following is not a fixed quantity
1) atomic weight of an element 2) equivalent weight of an element (or) compound
3) molecular weight of a compound 4) formula weight of a substance
9. At 25°C for a given solution $M = m$, then at 50°C the correct relationship is
1) $M = m$ 2) $M > m$ 3) $M < m$ 4) $M = 2M$
10. The units of molality are
1) moles, lit^{-1} 2) moles, ml^{-1} 3) moles, Kg^{-1} 4) g. equivalents Kg^{-1}
11. (A) : Molality is independent of temperature.
(R) : There is no volume factor in the expression of molality.
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false
4) (A) is false but (R) is true
12. The number of Glucose molecules present in 10 ml of decimolar solution is
1) 6.0×10^{20} 2) 6.0×10^{19} 3) 6.0×10^{21} 4) 6.0×10^{22}
13. The number of ions present in 1 ml of 0.1M CaCl_2 solution is
1) 1.8×10^{20} 2) 6.0×10^{20} 3) 1.8×10^{19} 4) 1.8×10^{21}
14. 100 ml of an aqueous solution contains 6.023×10^{21} solute molecules. The solution is diluted to 1 lit. The number of solute molecules present in 10ml of the dilute solution is
1) 6.0×10^{20} 2) 6.0×10^{19} 3) 6.0×10^{18} 4) 6.0×10^{17}
15. 0.1 gram mole of urea is dissolved in 100g. of water. The molality of the solution is
1) 1 m 2) 0.01 M 3) 0.01 m 4) 1.0 M
16. 6 g. of Urea is dissolved in 90 g. of water. The mole fraction of solute is
1) 1/5 2) 1/50 3) 1/51 4) 1/501

Numerical Value Type Questions

17. The number of millimoles of H_2SO_4 present in 5 litres of 0.2N H_2SO_4 solution is
18. 200 ml of 1M H_2SO_4 , 300 ml 3M HCl and 100 ml of 2M HCl are mixed and made upto 1 litre. The proton concentration in the resulting solution is
19. The Normality of 0.98% (w/v) H_2SO_4 solution is

LEVEL-II (ADVANCED)**Straight Objective Type Questions**

1. Which of the following differs from the others?
a) ruby b) elektron c) bell metal d) amalgam

2. H_2SO_4 is labelled as 9.8% by weight. Specific gravity of H_2SO_4 is 1.8. The volume of the acid to be taken to prepare 1000 ml of 0.18M solution is
a) 10ml b) 100ml c) 740 ml d) 360 ml
3. The volume of decamolar aqueous solution of HCl is required to prepare 2dm^3 of 5M HCl sol. is
a) 0.5L b) 1L c) 2L d) 3L
4. The molality of 2% (W/W) NaCl solution nearly
a) 0.02m b) 0.35 m c) 0.25 m d) 0.45 m
5. The mole percentage of oxygen in a mixture of 7 gm of Nitrogen and 8 gm of oxygen is
a) 8 b) 16 c) 21 d) 50
6. The volumes of 1M HCl and 5M HCl to be mixed to get 2 lit of 2M HCl are
a) 1 lit and 1 lit b) 1.5 lit and 0.5 lit c) 1.25 lit and 0.75 lit d) 1.33 lit and 0.66 lit
7. A solution weighing 'a' gm has molality 'b'. The molecular mass of solute. If the mass of solute is 'c' gm will be
a) $\frac{c}{b} \cdot \frac{1000}{(a-c)}$ b) $\frac{b}{a} \cdot \frac{1000}{(a-b)}$ c) $\frac{b}{c} \cdot \frac{1000}{(a-c)}$ d) $\frac{c}{a} \cdot \frac{1000}{(b-a)}$
8. 10 millimoles of a diacidic base exactly neutralises 100 ml of an acid. Then the N of that acid is
a) 0.2 N b) 0.1 N c) 0.4 N d) 0.5N
9. 100 mL of H_2SO_4 solution having molarity 1M and density 1.5g/ml is mixed with 400 ml of water. Calculate final molarity of H_2SO_4 solution, if final density is 1.25 g/ml :
a) 4.4 M b) 0.145 M c) 0.52 M d) 0.227 M
10. Wood's metal contains 50.0% bismuth, 25.0% lead, 12.5% tin and 12.5% cadmium by weight. What is the mole fraction of tin? (Atomic weights : Bi = 209, Pb = 207, Sn = 119, Cd = 112)
a) 0.202 b) 0.158 c) 0.176 d) 0.221
11. The density of a 56.0% by weight aqueous solution of 1-propanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$) is 0.8975 g/cm^3 . What is the mole fraction of the compound?
a) 0.292 b) 0.227 c) 0.241 d) 0.276
12. Three sols. X, Y, Z of HCl are mixed to produce 100 mL of 0.1 M sol.. The molarities of X, Y and Z are 0.07 M, 0.12 M and 0.15 M respectively. What respective vol. of X, Y and Z should be mixed?
a) 50 ml, 25 ml, 25 ml b) 20 ml, 60 ml, 20 ml
c) 40 ml, 30 ml, 30 ml d) 55 ml, 20 ml, 25 ml

More than One correct answer Type Questions

13. The characteristic property of solution is
a) formation of solution is physical change
b) solute and solvent in the solution can be separated by filtration
c) solute and solvent in the solution can be separated by distillation
d) solution can be represented with a chemical formula
14. In a solution 49 gm of H_2SO_4 is dissolved in 2 litre of solution. The concentration of solution may be
a) 0.5 M b) 0.25 M c) 0.5 N d) 0.25 N
15. 5.3% (w/v) Na_2CO_3 solution and 6.3% (w/v) $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ solution have same
a) molality b) molarity c) normality d) mole fraction

16. Which of the following concentration terms is added by a change of in temperature?
 a) Molarity b) Molality c) Normality d) Specific Gravity

Linked Comprehension Type Questions*Passage:*1.24M aqueous solution of KI has density 1.15 g/cm³

17. Percentage composition of solute in the solution is
 a) 17.89 b) 27.89 c) 37.89 d) 47.89
18. Molality of this solution is
 a) 2.61 b) 1.31 c) 4.12 d) 3.12
19. Normality of this solution is
 a) 0.62 b) 1.24 c) 2.48 d) 3.72

Matrix Matching Type Questions20. **Column-I**

- A) Molarity
 B) Molality
 C) Formality
 D) Strength of solution

Column-II

- p) Number of gram formula mass of solute dissolved per litre of solution
 q) Number of moles of solute dissolved per kg of solvent
 r) Depends on Temperature
 s) Number of moles of solute dissolved per litre of solution

21. **Column-I**

- A) Mole fraction
 B) Molality
 C) Molarity
 D) Formality

Column-II

- p) mole/kg
 q) Depends on temperature
 r) No units
 s) Independent on temperature

Integer Type Questions

22. What volume of liquid (in lit) will contain 10moles? If molar mass of liquid is 280 and its density is 1.4 gm/ml?
23. 200ml of 1M HCl is mixed with 300ml of 6M HCl and the final solution is diluted to 1000ml. Calculate molar concentration of [H⁺] ion.

EXERCISE-II**(Vapour Pressure, Raoult's Law, Ideal & Non-ideal solutions, Henry's Law, Azeotropic Mixture)****LEVEL-I (MAIN)****Straight Objective Type Questions**

1. Which one of the following gases has the lowest value of Henry's Law constant?
 1) N₂ 2) He 3) CO₂ 4) O₂
2. The solubility of N₂(g) in water exposed to the atmosphere, when the partial pressure of 593 mm is 5.3×10^{-4} M. Its solubility at 760 mm and at the same temperature is:
 1) 4.1×10^{-4} M 2) 6.8×10^{-4} M 3) 1500 M 4) 2400 M
3. As temperature increases, vapour pressure of a liquid
 1) increases linearly 2) decreases linearly
 3) increases exponentially 4) decreases exponentially

4. Rate of evaporation depends upon
 a) Nature of liquid
 b) Surface area of the liquid
 c) Temperature
 d) Flow of air current over the surface
 The correct answer is
 1) a, b only 2) b, c only 3) a, b, and c only 4) a, b, c and d
5. (A) : At boiling point of the liquid vapour pressure equal to atmospheric pressure
 (R) : Volume of a solution increases by increasing the temperature.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
6. (A) : Sea water boils at higher temperature than distilled water.
 (R) : Addition of non volatile solute to a solvent lowers the vapour pressure
7. (A) : A pressure cooker reduces cooking time.
 (R) : The boiling point of water inside the cooker is increased.
8. A solution that obeys Raoult's law is called
 1) normal solution 2) non-ideal solution 3) ideal solution 4) saturated solution
9. At 20°C, the vapour pressure of diethyl ether is 442mm. When 6.4 g. of a non-volatile solute is dissolved in 50g. of ether, the vapour pressure falls to 410 mm. The Molecular weight of the solute is
 1) 150 2) 130.832 3) 160 4) 180
10. Among the following the azeotropic mixture is
 1) $\text{CCl}_4 + \text{CHCl}_3$ 2) $\text{C}_6\text{H}_{14} + \text{C}_7\text{H}_{16}$ 3) $\text{C}_2\text{H}_5\text{Br} + \text{C}_2\text{H}_5\text{Cl}$ 4)  + 
11. An azeotropic solution of two liquids has a boiling point lower than either of them when it
 1) shows negative deviation from Raoult's law 2) shows no deviation from Raoult's law
 3) shows positive deviation from Raoult's law 4) is saturated

Numerical Value Type Questions

12. 6 g. of urea is dissolved in 90 g. of boiling water. The vapour pressure of the solution is ____ mm.
13. The vapour pressure of pure water at 25°C is 30 mm. The vapour pressure of 10% (W/W) glucose solution at 25°C is ____ mm.
14. The wt. of urea to be dissolved in 100g. of water to decrease the vapour pressure of water by 5% is ____ g
15. The vapour pressure of a pure liquid A is 40 mm of Hg at 310K. The vapour pressure of this liquid in a solution with liquid B is 32 mm of Hg. x_A in the solution, if it obeys Raoult's law, is :

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. (A) : If one component obeys Raoult's law over a certain range of composition, the other component would not obey Henry's law in that range
 (R) : Raoult's law is a special case of Henry's law.
 a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 c) (A) is true but (R) is false d) (A) is false but (R) is true

2. 139.18 g of glucose is added to 178.2 g of water the vapour pressure of water for this aqueous solution at 100°C is
 a) 700.7 torr b) 759 torr c) 7.6 torr d) 76 torr
3. The relative lowering of vapour pressure of 0.2 molal solution in which the solvent is Benzene is
 a) 15.6×10^{-4} b) 15.6×10^{-3} c) 15.6×10^{-1} d) 0.05
4. Vapour pressure of an aqueous solution is 2% less than that of the solvent. The molality of the solution is
 a) 2m b) 1.5 m c) 1.13 m d) 0.2 m
5. The amount of Glucose to be dissolved in 500 g. of water so as to produce the same lowering in vapour pressure as that of 0.2 molal aqueous urea solution
 a) 9 g. b) 18 g. c) 36 g. d) 1.8 g.
6. A Current of dry air was first passed through the bulb containing solution of 'A' in water and then through the bulb containing pure water. The loss in mass of a solution bulb is 1.92g gm Where as that in pure water bulb is 0.08g, then mole fraction of 'A' is
 a) 0.86 b) 0.2 c) 0.96 d) 0.04
7. The vapour pressure of a mixture of 1 mole of liquid 'A' and 3 mole of liquid 'B' is 550 mm of Hg. When one mole of liquid 'B' is further added, the vapour pressure of the solution increases by 10 mm of Hg. The vapour pressure of pure liquid A and B are respectively in mm of Hg.
 a) 400, 600 b) 200, 800 c) 500, 600 d) 600, 800
8. The vapour pressure of pure benzene and toluene are 160 Torr and 60 Torr respectively. The mole fraction of benzene in vapour phase in contact with equimolar solution of benzene and toluene would be
 a) 0.50 b) 8/3 c) 0.73 d) 0.27
9. The plot of $1/y_A$ versus $1/y_A$ (where x_A and y_A are the mole fraction of A in liquid and vapour phases, respectively) is linear with slope and intercept respectively are given as (y-axis = $1/y_A$, x-axis = $1/x_A$)
 a) $P_A^0/P_B^0, (P_A^0 - P_B^0)/P_B^0$ b) $P_A^0/P_B^0, (P_B^0 - P_A^0)/P_B^0$
 c) $P_B^0/P_A^0, (P_A^0 - P_B^0)/P_A^0$ d) $P_B^0/P_A^0, (P_B^0 - P_A^0)/P_B^0$
10. An azeotropic solution of two liquids has a boiling point higher than either of the boiling points of the liquids when it
 a) shows negative deviation from Raoult's law b) shows positive deviation from Raoult's law
 c) shows no deviation from Raoult's law d) is saturated
11. Solution (A) is ideal and solution (B) is non-ideal with -ve deviation, which of the following are correct?
 a) $\Delta H_{\text{mix}} = 0, \Delta V_{\text{mix}} = 0$ for A b) ΔS_{mix} is zero for both A, B
 c) ΔH_{mix} is positive, ΔV_{mix} is negative for B d) ΔH_{mix} is negative, ΔV_{mix} is positive for B

More than One correct answer Type Questions

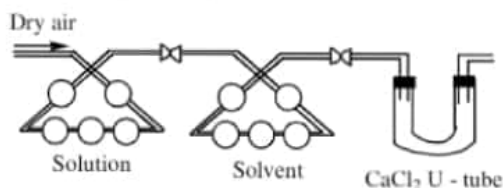
12. At a given temperature, correct statements:
 a) Vapour pressure of a sol. containing nonvolatile solute is proportional to mole fraction of solvent
 b) Lowering of vapour pressure of solution containing nonvolatile solute is proportional to mole fraction of solute
 c) Relative lowering of vapour pressure is equal to mole fraction of solute
 d) Vapour pressure of a sol. containing nonvolatile solute is proportional to mole fraction of solute

13. Which of the following statements are correct?
- the boiling point of a solution is greater than that of pure solvent
 - the temperature where the V.P. of liquid equals to atmospheric pressure is called its boiling point
 - the V.P. of pure solvent is less than the vapour pressure of solution containing non volatile solute.
 - the T of liquid remained in the container after evaporation is more than that before the evaporation
14. For a solution containing nonvolatile solute, the relative lowering of vapour pressure is 0.2, If total moles present in the solution are 5 then select the correct relations
- $x_{\text{solute}} = 0.2$
 - $n_{\text{solute}} = 1$
 - $n_{\text{solvent}} = 4$
 - $x_{\text{solvent}} = 0.8$
15. Which of the following show deviation from Raoult's law?
- $\text{C}_2\text{H}_5\text{OH}$ & H_2O
 - HNO_3 & H_2O
 - CHCl_3 & CH_3COCH_3
 - C_6H_6 & $\text{C}_6\text{H}_5\text{CH}_3$
16. 4 mole of pure liquid A ($P_A^0 = 80\text{mm of Hg}$) and 6 mole of pure liquid B ($P_B^0 = 100\text{mm of Hg}$) are mixed. The vapour pressure of the resulting solution is found to be 90 mm of Hg then
- It is an ideal solution
 - The solution shows negative deviation from Raoult's law
 - The solution is formed by the absorption of heat
 - The solution will boil at higher temperature than expected
17. Which of the following confirm minimum boiling point azeotropic mixture?
- $\text{CCl}_4 + \text{CHCl}_3$
 - $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{N}$
 - $\text{CH}_3\text{COCH}_3 + \text{CHCl}_3$
 - $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O}$
18. At certain temperature a solution 3 moles of liquid A ($p^0 = 500$ torr) and 1 mol of liquid B ($p^0 = 340$ torr) has a vapour pressure of 435 torr, we can conclude that
- a solution of 50 mL of A and 50 mL of B must have a volume of 100 mL.
 - the solution shows negative deviation from Raoult's law
 - on mixing the two liquids heat must be given out so as to obtain solution at the same temperature
 - entropy of mixing is zero.

Linked Comprehension Type Questions

Passage-I :

A current of dry air is passed through the above apparatus containing two sets of bulbs. One set of bulbs is filled with the solution and second with the solvent. Finally that is attached with a U-tube containing CaCl_2 , which is very efficient in absorbing the water vapour. Passing of dry air through all these three causes some change in weight.

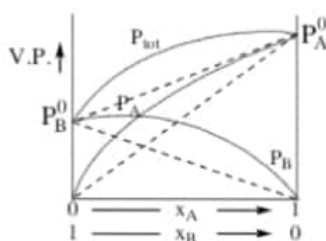


19. Loss in weight (W_1) of solution is proportional to
 a) the weight of air passed in
 b) the vapour pressure of solution
 c) the vapour pressure of pure solvent
 d) the weight of solute
20. Loss in weight (W_2) of solvent (water) is proportional to
 a) the vapour pressure of solution
 b) the vapour pressure of pure solvent
 c) the difference $P^0_{\text{solvent}} - P_{\text{solution}}$
 d) the mol fraction of solute
21. Gain in weight (W_3) of U-tube containing CaCl_2 is proportional to
 a) the vapour pressure of solution
 b) the vapour pressure of pure solvent
 c) the difference $P^0_{\text{solvent}} - P^0_{\text{solution}}$
 d) the mol fraction of solute

Passage-II:

Vapour Pressure of a mixture of benzene and toluene is given by $P = 179X_B + 92$, Where X_B is molefraction of benzene.

22. Vapour pressure of the solution obtained by mixing 936 g of benzene and 736 gm of toluene
 a) 199.4 mm b) 271 mm c) 280 mm d) 289 mm
23. If Vapours are removed and condensed into liquid then what will be the ratio of mole fraction of benzene and toluene in first condensate
 a) 2.8 b) 1.5 c) 3.5 d) 4.5
24. This condensed liquid again brought to the same temperature then what will be the mole fraction of benzene in vapour phase
 a) 0.07 b) 0.93 c) 0.65 d) 4.5

Passage-III :

25. Consider some facts and select the correct facts.
 I) This is observed when A....B attractions are greater than average of A....A and B....B attraction.
 II) $\Delta H_{\text{mix}} = +ve$, $\Delta V_{\text{mix}} = +ve$
 III) Boiling point is smaller than expected such that vaporisation is increased
 IV) Mixture is called Azeotropic mixture
 a) I, II, III b) II, III, IV c) I, III, IV d) I, II, III, IV
26. Total vapour pressure of mixture of 1 mole of volatile component A ($P_A^0 = 100 \text{ mm Hg}$) and 3 mole of volatile component B ($P_B^0 = 60 \text{ mm Hg}$) is 75 mm. For such case
 a) There is +ve deviation from Raoult's law
 b) Boiling point has been lowered
 c) Force of attraction between A and B is smaller than that between A and A or between B and B
 d) All the above statements are correct

Matrix Matching Type Questions

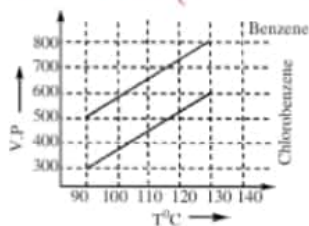
27. Column-I

- A) Lowering of vapour pressure
 B) Relative lowering of vapour pressure
 C) Raoult's law
 D) Ideal solution

Column-II

- p) $\frac{P^0 - P}{P^0}$
 q) $\frac{P^0 - P}{P^0} = \frac{w}{m} \times \frac{M}{W}$
 r) $P^0 - P$
 s) Obeying Raoult's law
 t) Boiling point

28. Using V.P Vs temp graph match the following : $\left(X_{\text{chlorobenzene}} = \frac{1}{3}, X_{\text{Benzene}} = \frac{2}{3} \right)$



Column-I

- A) P at 90°
 B) $X_{\text{chlorobenzene}}$ in vapour phase
 C) X_{benzene} in vapour phase at 130°
 D) P at 130°

Column-II

- p) 0.23
 q) 0.73
 r) 733.33
 s) 433.33

29. Column-I (graph)

- A)
 B)
 C)
 D)

Column-II (suitable variables)

- p) Vapour pressure vs. composition
 q) Boiling point of non-ideal mixture vs. composition
 r) $\log(\text{V.P.})$ vs. $(\text{temperature})^{-1}$
 s) Vapour pressure vs. temperature.

30. Column-I

- A) n-hexane, n-heptane
 B) CO_2 , H_2O
 C) $\text{C}_6\text{H}_5\text{COOH}$, Benzene
 D) $\text{C}_6\text{H}_5\text{COOH}$, H_2O

Column-II

- p) Dimerisation
 q) ionisation
 r) Raoult's law (obeys)
 s) Henry's law (obeys)

Integer Type Questions

31. $P_A = 320$ torr, when $x_A = x$ torr. If $x_A = x + 0.1$; $P_A = 440$ then $P_A^0 = 200x$; find x .
 32. $P_A^0 = 10$ torr; 1g of B is dissolved in 20g of A, its vapour pressure reduced to 9.0 torr. If M.W of A is 200, M.W of B is $30x$. x is

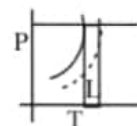
EXERCISE-III

(ΔT_b , ΔT_f , π , Van't Hoff's factor)

LEVEL-I (MAIN)

Straight Objective Type Questions

- During depression of freezing point in a solution the following are in equilibrium
 - Liquid solvent, Solid solvent
 - Liquid solvent, Solid solute
 - Liquid solute, Solid Solute
 - Liquid solute, Solid solvent
- The molal elevation constant is the ratio of the elevation in boiling point to
 - Molarity
 - Molality
 - Mole fraction of solute
 - Mole fraction of solvent
- Molality of an aqueous solution that produces an elevation of boiling point of 1.00 K at 1 atm pressure. (K_b for water = $0.512 \text{ K} \cdot \text{kg} \cdot \text{mol}^{-1}$)
 - 0.512 M
 - 0.195 m
 - 1.95 m
 - 5.12 M
- Which of the following is a colligative property?
 - vapour pressure of a liquid
 - boiling point
 - freezing point
 - relative lowering of vapour pressure of a sol.
- The phase diagrams for the pure solvent (solid lines) and the solution (non-volatile solute, dashed line) are recorded below.
 The quantity indicated by 'L' in the figure is
 - Δp
 - ΔT_f
 - $K_b \cdot m$
 - $K_f \cdot m$
- An aqueous solution containing one gram of urea boils at 100.25°C . The aqueous solution containing 3gm of glucose in the same volume will boil at
 - 100°C
 - 100.25°C
 - 100.5°C
 - 100.75°C
- Y g of a non-volatile Organic substance of molecular mass M is dissolved in 250g benzene Molal elevation constant of benzene is k_b . Elevation in its B.P is given by
 - $\frac{M}{K_b Y}$
 - $\frac{4K_b Y}{M}$
 - $\frac{K_b Y}{4M}$
 - $\frac{K_b Y}{M}$



8. The solution containing 6.8g of non-ionic solute in 100g of water was found to freeze at -0.93°C . If K_f for water is 1.86, the molar mass of solute is
 1) 13.6 2) 68 3) 34 4) 136
9. Solution S_1 contains 3g of urea per litre and solution S_2 contains 9g glucose per litre. At 298 K, the osmotic pressure of
 1) S_1 is greater than that of S_2 2) S_1 is less than that of S_2
 3) Both the solution is same 4) Both the solution is 1 atm
10. The Osmotic pressure of solution containing 4.0g of solute (molar mass 246) per litre at 27°C is ($R = 0.082\text{L atm K}^{-1}\text{mol}^{-1}$):
 1) 0.1 atm 2) 0.2 atm 3) 0.4 atm 4) 0.8 atm
11. Which of the following colligative properties can provide molar mass of proteins (or polymers or colloids) with greater precision?
 1) Relative lowering of vapour pressure 2) Elevation of boiling point
 3) Depression in freezing point 4) Osmotic pressure
12. The degree of dissociation (α) of a weak electrolyte A_xB_y is related to Van't Hoff factor (i) by the expression
 1) $\alpha = \frac{i-1}{(x+y-1)}$ 2) $\alpha = \frac{i-1}{x+y+1}$ 3) $\alpha = \frac{x+y-1}{i-1}$ 4) $\alpha = \frac{x+y+1}{i-1}$
13. If α is the degree of dissociation of Na_2SO_4 , the Vant Hoff factor (i) used for calculating the molecular mass is
 1) $1+\alpha$ 2) $1-\alpha$ 3) $1+2\alpha$ 4) $1-2\alpha$
14. Relative lowering of vapour pressure is maximum for
 1) 0.1m urea 2) 0.1m NaCl 3) 0.1m MgCl_2 4) 0.1m $\text{Al}_2(\text{SO}_4)_3$
15. Boiling point is least for
 1) 0.1m urea 2) 0.2m urea 3) 0.1m NaCl 4) 0.2m MgCl_2
16. (A) : Vapour pressure of 0.5M sugar solution is more than 0.5M KCl solution
 (R) : Lowering of vapour pressure is directly proportional to the number of particles present in the solution.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
17. Solution A, B, C and D are respectively 0.1M Glucose, 0.05M NaCl, 0.05M BaCl_2 and 0.1M AlF_3 . Which one of the following pairs are isotonic?
 1) A & C 2) B & C 3) A & B 4) A & D
18. Solution A contains 7g/L MgCl_2 and solution B contains 7g/L of NaCl. At room temperature, the osmotic pressure of
 1) solution A is greater than B 2) both have same osmotic pressure
 3) solution B is greater than A 4) Can't determine
19. Which has the highest boiling point?
 1) 0.1M Na_2SO_4 2) 0.1M $\text{C}_6\text{H}_{12}\text{O}_6$ 3) 0.1M MgCl_2 4) 0.1M $\text{Al}(\text{NO}_3)_3$

20. 0.004M Na_2SO_4 is isotonic with 0.01M glucose. Degree of dissociation of Na_2SO_4 is :

- 1) 75% 2) 50% 3) 25% 4) 85%

21. When mercuric iodide is added to the aqueous solution of potassium iodide, the

- 1) freezing point is raised 2) freezing point does not change
3) freezing point is lowered 4) boiling point does not change

Numerical Value Type Questions

22. A solution prepared by dissolving 0.8gm of naphthalene in 100g of CCl_4 has a boiling point elevation of 0.4°C . A 1.24 g of an unknown solute in same amount of CCl_4 produced boiling point elevation of 0.62°C , then molar mass of unknown solute is

23. The molal freezing point constant for water is $1.85 \text{ K.kg mole}^{-1}$. The freezing point of 0.1m NaCl solution is $^\circ\text{C}$

24. Average osmotic pressure of human blood is 7.4 atm at 27°C , then total concentration of various solutes is

25. Consider equimolal aqueous solutions of NaHSO_4 and NaCl with ΔT_b and ΔT_b^1 as their respective boiling point elevations. The value of $\lim_{v \rightarrow \infty} \frac{\Delta T_b}{\Delta T_b^1}$ will be

26. 0.6 molar MgCl_2 and 1.2 molar sucrose sol. are isotonic then the percentage degree of dissociation of MgCl_2 is

27. Lowering of vapour pressure of 1.00 m solution of a non-volatile solute in a hypothetical solvent of molar mass 40g at its normal boiling Points is torr .

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. For a dilute solution containing 2.5 g of non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C . Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution is (take $K_b = 0.76 \text{ kg mole}^{-1}$)

- a) 724 b) 740 c) 736 d) 718

2. Molal depression constant for water is $1.86 \text{ K.Kg.mole}^{-1}$. The freezing point of a 0.05 molal solution of a non electrolyte in water is

- a) -1.86°C b) -0.93°C c) -0.093°C d) 0.93°C

3. The relationship between osmotic pressure at 273K when 10g glucose (P_1), 10g urea (P_2) and 10 g sucrose (P_3) are dissolved in 250 ml of water is

- a) $P_1 > P_2 > P_3$ b) $P_3 > P_1 > P_2$ c) $P_2 > P_1 > P_3$ d) $P_2 > P_3 > P_1$

4. If 0.1 M solution of glucose and 0.1 M solution of urea are placed on two sides of a semi-permeable membrane to equal heights, then it will be correct to say that

- a) Glucose will flow towards urea solution
b) There will be no net movement across membrane
c) Urea will flow towards glucose solution
d) Water will flow from urea solution to glucose

5. Set-I

- i) RBC in 0.5% NaCl solution
- ii) RBC in 1% NaCl solution
- iii) egg (outer shell removed in water)
- iv) egg (outer shell removed in NaCl solution)

Incorrect match is

a) i - A

b) ii - B

Set-II

- A) Swells
- B) Shrinks

c) iii - A

d) iv - A

6. Which of the following statement is false?

- a) Raoult's law states that vapour pressure of a component over a solution is proportional to its mole fraction.
- b) Osmotic pressure is given by the expression $\pi = MRT$ where, M is molarity.
- c) The correct order of osmotic pressures of 0.01 M aqueous solution of each compound is $\text{BaCl}_2 > \text{KCl} > \text{CH}_3\text{COOH} > \text{Sucrose}$.
- d) Two sucrose solutions of same molarity prepared in different solvents will have same freezing point depressions.

7. Equimolar solutions of electrolytes in the same solvent have

- a) Same B.P. but different F.P.
- b) Same F.P. but different B.P.
- c) Same boiling and same freezing points
- d) Different boiling and different freezing point

8. Which of the following salt will have same value of Van't Hoff's factor [i] as that of $\text{K}_4[\text{Fe}(\text{CN})_6]$

- a) $\text{Al}_2(\text{SO}_4)_3$
- b) NaCl
- c) $\text{Al}(\text{NO}_3)_3$
- d) Na_2SO_4

9. Blood is isotonic with

- a) 0.16 M NaCl
- b) Conc. NaCl
- c) 50% NaCl
- d) 30% NaCl

10. The Osmotic pressure of blood is 8.21 atm at 37°C . How much glucose is used per lit for an intravenous injection that is isotonic with blood.

- a) 180 gm
- b) 342 gm
- c) 58.06 gm
- d) 55.55 gm

11. If M_{normal} is the normal molecular mass and α is the degree of ionisation of $\text{K}_3[\text{Fe}(\text{CN})_6]$, then the abnormal molecular mass of the complex in the solution will be

- a) $M_{\text{normal}} (1+2\alpha)^{-1}$
- b) $M_{\text{normal}} (1+3\alpha)^{-1}$
- c) $M_{\text{normal}} (1+\alpha)^{-1}$
- d) equal to M_{normal}

12. 1.0 molal aqueous solution of an electrolyte X_3Y_2 is 25% ionized. The Boiling point of the solution is (k_b of $\text{H}_2\text{O} = 0.52 \text{ K kg mol}^{-1}$)

- a) 375.5 K
- b) 374.04 K
- c) 377.12 K
- d) 373.25 K

13. pH of a 0.1M monobasic acid is found to be 2. Hence, its osmotic pressure at a given temp.

- a) 0.1RT
- b) 0.11RT
- c) 1.1RT
- d) 0.01RT

14. 4g of Substance A, dissolved in 100g H_2O depressed the freezing point of water by 0.1°C . While 4g of another substance B, depressed the f.pt by 0.2°C . What is the relation between molecular weights of A and B.

- a) $M_A = 4M_B$
- b) $M_A = M_B$
- c) $M_A = 0.5 M_B$
- d) $M_A = 2M_B$

15. The osmotic pressure of solution containing non-volatile, non-electrolyte solute is 7.6 atm at 27°C then the boiling point of the same solution is (Assume that molality and molarity are same)

- a) 126.25°
- b) 50.8°
- c) 100.16°
- d) 200.32°

More than One correct answer Type Questions

16. In the depression of freezing point experiment, correct statements are:
 a) The vapour pressure of the solution is less than that of pure solvent
 b) The vapour pressure of the solution is more than that of pure solvent
 c) Only solute molecules solidify at the freezing point
 d) Only solvent molecules solidify at the freezing point
17. Cryoscopic constant value depends on
 a) molar mass of solvent in the solution
 b) molar mass of solute in the solution
 c) the enthalpy of vapourisation of solvent
 d) freezing point of solvent
18. When a solution containing nonvolatile solute is diluted with water
 a) Its vapour pressure increases
 b) Its osmotic pressure decreases
 c) Its boiling point decreases
 d) Its freezing point increases

Linked Comprehension Type Questions

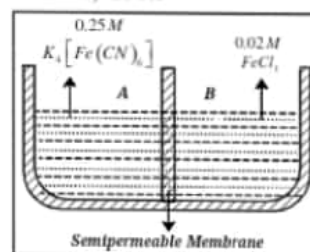
Passage-I :

The addition of 3 gm of substance to 100 gm CCl_4 ($M = 154 \text{ gm mol}^{-1}$) raises the boiling point of CCl_4 by 0.6°C . $K_b(\text{CCl}_4)$ is $5.03^\circ\text{C kg mol}^{-1}$, $k_f(\text{CCl}_4)$ is $31.8^\circ\text{C kg mol}^{-1}$ and density of solution is 1.64 gm/ml .

19. Calculate the depression in freezing point
 a) 3.79°C
 b) 0.018°C
 c) 4.65°C
 d) 251.5°C
20. The Osmotic pressure at 298K is :
 a) 3.79 atm
 b) 0.018 atm
 c) 4.65 atm
 d) 251.5 atm
21. The molar mass of the substance is :
 a) 37.9
 b) 18
 c) 46.5
 d) 251.5

Passage-II :

FeCl_3 on reaction with $\text{K}_4\text{Fe}(\text{CN})_6$ in aqueous solution gives a blue colour. These two solutions are separated by a semipermeable membrane as shown in the figure. Both the solutions are sufficiently dilute & osmosis will take place as there is a difference in concentration of the solutions. Imagine the experiment is carried out at 27°C . The osmotic flow will take place in one direction and not in both.



22. Which of the following observations may be recorded after a while?
 a) No change in colour in either of the compartments
 b) Blue colour appears in compartment A
 c) Blue colour appears in compartment B
 d) Blue colour appears in both A and B
23. In the given experiment, with passage of time, the concentrations are expected to change as
 a) Concentration of FeCl_3 Decreases
 b) Concentration of $\text{K}_4\text{Fe}(\text{CN})_6$ Increases
 c) Concentration of FeCl_3 Increases
 d) Concentration of Both Remain Unchanged
24. The osmotic pressure (π) of FeCl_3 Solution can be expressed as
 a) 6 R
 b) 18 R
 c) 24 R
 d) 5 R

Passage-III :

$\text{CoCl}_3 \cdot x\text{NH}_3$ is a complex compound obtained by dissolving CoCl_3 in ammonia. The complex compound is denoted as $[\text{Co}(\text{NH}_3)_6-x]\text{Cl}_{3-x}$ (soluble 100% ionisable). The chlorides shown inside the square brackets are not ionisable.

The salt ionises as $[\text{Co}(\text{NH}_3)_{6-x}\text{Cl}_x]\text{Cl}_{3-x} \rightarrow [\text{Co}(\text{NH}_3)_{6-x}\text{Cl}_x]^{+(3-x)} + (3-x)\text{Cl}^-$

25. 0.1M " $\text{CoCl}_3 \cdot x\text{NH}_3$ " solution has same osmotic pressure as 0.1M glucose solution. What is 'n' ?
a) 2 b) 3 c) 4 d) 5
26. 0.1m $\text{CoCl}_3 \cdot x\text{NH}_3$ solution is found to boil at 100.104°C (K_b of water = 0.52K Kg mol^{-1}) what x? (α of salt = 1)
a) 2 b) 3 c) 4 d) 5
27. 66% of chloride content of the complex $\text{CoCl}_3 \cdot x\text{NH}_3$ can be precipitated by excess of aq. AgNO_3 . What is the Van't Hoff factor of the complex for 50% ionisation?
a) 2 b) 3 c) 4 d) 2.5

Matrix Matching Type Questions**28. Column-I**

- A) π_1 : 0.1M NaCl ; π_2 : 0.1M urea
B) π_1 : 0.1M NaCl ; π_2 : 0.1M Na_2SO_4
C) π_1 : 0.1M NaCl ; π_2 : 0.1M KCl
D) π_1 : 0.1M CuSO_4 ; π_2 : 0.1M sucrose

Column-II

- p) π_1 is isotonic to π_2
q) No net migration of solvent across the membrane
r) π_1 is hypertonic to π_2
s) π_1 is hypotonic to π_2

29. Column-I

- A) 0.1M $\text{Al}_2(\text{SO}_4)_3$
B) 0.1M H_3PO_4
C) 0.1 M urea
D) 0.1 M MgCl_2

Column-II

- p) Solution with highest boiling point
q) Van't Hoff factor is greater than 1
r) Solution with lowest osmotic pressure
s) Solution with lowest freezing point

30. Match with isotonic solutions.**Column-I**

- A) 6% w/v urea ($M=60$) aqueous solution
B) 5.85% w/v NaCl
C) 0.2 M NH_4Cl aq. solution ($\alpha=1$)
D) 0.2 M $\text{K}_3[\text{Fe}(\text{CN})_6]$ aq solution ($\alpha=1$)

Column-II

- p) 0.5M K_2SO_4 (aq) solution ($\alpha=1$)
q) 0.1M $\text{K}_3[\text{Fe}(\text{CN})_6]$ (aq) solution ($\alpha=1$)
r) 36% w/v glucose (aq) solution
s) 18% w/v glucose(aq) solution

Integer Type Questions

31. 718.2 g of sucrose is dissolved in 1 kg of water. If '100+x' is the difference between the melting and boiling point of the solutions, then report the value of 'x'.
32. The boiling point of an aqueous solution is found to be 100.28°C . Then the freezing point of the same solution is $-x^\circ\text{C}$. What is the value of 'x' ?
33. A solution of a non-volatile solute in ethanol (B.P of ethanol = 78.4°C) has a vapour pressure of 730mm of Hg at 78.4°C . To what temperature, the above solution must be heated have the pressure of 760mm of Hg? K_b of ethanol = 1.22K Kg mol^{-1} . Express temperature of $10x$. x is ____
34. For $[\text{CrCl}_3 \cdot x\text{NH}_3]$, elevation in boiling point of one molal solution is double of one molal urea solution. Hence, the value of x (assuming complete dissociation) is ____
35. Mole fraction of solute in some solution is $1/n$. If 50% of solute molecules dissociate into two parts and remaining 50% get dimerised, new mole fraction of solvent becomes $4/5$. Find value of n.

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

- 1) 1 2) 3 3) 3 4) 2 5) 4 6) 1 7) 3 8) 2
 9) 3 10) 3 11) 1 12) 1 13) 1 14) 2 15) 1 16) 3
 17) 500 18) 1.5 19) 0.2

LEVEL-II

- 1) a 2) b 3) b 4) b 5) d 6) b 7) a 8) a
 9) d 10) c 11) d 12) d 13) ac 14) bc 15) bcd 16) acd
 17) a 18) b 19) b 20) A-rs; B-q; C-pr; D-r
 21) A-rs; B-ps; C-q; D-q 22) 2 23) 2

EXERCISE-II

LEVEL-I

- 1) 3 2) 2 3) 3 4) 4 5) 2 6) 1 7) 1 8) 3
 9) 2 10) 1 11) 3 12) 744.813) 29.6714) 16.66 15) 0.8

LEVEL-II

- 1) d 2) a 3) b 4) c 5) b 6) d 7) a 8) c
 9) c 10) a 11) a 12) abc 13) ab 14) abcd 15) abc 16) bd
 17) ad 18) bc 19) b 20) c 21) b 22) a 23) d 24) b
 25) b 26) d 27) A-r; B-p; C-q; D-s 28) A-s; B-p; C-q; D-r
 29) A-s; B-r; C-q; D-p 30) A-r; B-s; C-p; D-q
 31) 6 32) 3

EXERCISE-III

LEVEL-I

- 1) 1 2) 2 3) 3 4) 4 5) 3 6) 2 7) 2 8) 4
 9) 3 10) 3 11) 4 12) 1 13) 3 14) 4 15) 1 16) 1
 17) 3 18) 3 19) 4 20) 1 21) 1 22) 128 23) -0.37 24) 0.3
 25) 1.5 26) 50 27) 29.23

LEVEL-II

- 1) a 2) c 3) c 4) b 5) d 6) d 7) d 8) a
 9) a 10) c 11) b 12) b 13) b 14) d 15) c 16) ad
 17) acd 18) abcd 19) a 20) c 21) d 22) a 23) c 24) c
 25) b 26) c 27) a 28) A-r; B-s; C-pq; D-r
 29) A-pqs; B-q; C-r; D-q 30) A-s; B-r; C-q; D-p 31) 5 32) 1
 33) 8 34) 4 35) 6